PROJECT REPORT

UniNexus

An AI Academic Copilot for Smart University Management

TECHNOLOGY STACK

Laravel 11 · Vue 3 + Inertia 2 · Vite + Tailwind · MySQL 8

RAG Pipeline · Agentic Tool Calling · Layered AI Proctoring

Pluggable AI Providers — OpenAI · OpenRouter · Jina · Mock (keyless)

One platform. Grounded AI. Every answer, provable.

1. Executive Summary

UniNexus is an intelligent, centralised academic platform that unifies the fragmented digital life of a modern university into a single, role-aware system powered by AI. Rather than forcing students, faculty and administrators across ERP portals, messaging groups, classroom apps, spreadsheets and paper records, UniNexus brings academic communication, course management, attendance, assessments, document knowledge, analytics and AI-assisted decision support together in one place — one login, three tailored experiences.

Its defining innovation is a **Retrieval-Augmented Generation (RAG)** architecture. Unlike generic chatbots that hallucinate, UniNexus grounds every answer in the institution's own approved documents — syllabi, policies, handbooks, lecture materials — and in the user's own notes and their sections' course materials. Answers stream token by token and carry citations (down to the document and page) plus a confidence score. The result is trustworthy, auditable and safe for high-stakes academic use.

The platform has moved beyond answering to **acting**. The student tutor is agentic: through eight permission-checked tools it checks deadlines, lists courses, browses and books faculty office hours, generates practice quizzes and flashcard decks, and creates planner tasks — with a live tool-activity trail in the conversation and a server-enforced Agent / Answers-only switch. Every tool call runs through the same domain services, RBAC and section scoping as a UI click.

Assessment integrity has become a first-class capability. Class tests can be protected by **seventeen independent proctoring layers** — from fullscreen enforcement and randomisation to webcam/screen recording and three on-device camera-AI layers: a blink-verified face-liveness gate, phone and second-face detection, and snapshot photo evidence that replaces hours of video with moment-based photo bursts. Every attempt yields an evidence trail and a 0–100 risk-scored review dossier; every signal is a review aid for humans, never an automated verdict.

The AI itself is now **resilient by construction**. Four pluggable providers — OpenAI (gpt-4o + embeddings), OpenRouter (an OpenAI-compatible chain of free models), Jina (free multilingual embeddings) and a deterministic keyless Mock provider — sit behind one contract. Chat and embeddings each resolve through fallback chains that terminate in the never-failing Mock provider, vectors are model-tagged so providers never cross-compare, and switching providers automatically re-embeds the knowledge base. The entire product can be demonstrated on any laptop with no API keys, no vector database and no setup.

Finally, UniNexus is **operable as a real deployment**: an admin User Activity panel reports visitors, referrers, devices and IP-derived locations with zero request-path latency; a demo-mode cap meters AI requests per account across every AI surface so a public demo cannot drain the provider budget; and a hidden URL-driven maintenance switch can freeze or restore the whole platform instantly, with the health check always reachable.

At a glance

• Three role-based experiences (Student, Faculty, Admin) behind one login. • Streaming, grounded AI tutor with citations and confidence on every answer — from the library and from your own notes and materials. • Agentic in-chat actions: 8 permission-checked tools, live activity trail, hard server-side Answers-only mode. • 17-layer exam proctoring incl. on-device camera AI (face liveness, phone detection, snapshot evidence). • MySQL-native vector search — no separate vector database to deploy, operate or pay for. • Pluggable AI providers with automatic fallback: OpenAI, OpenRouter, Jina and a keyless Mock. • Production-ready core: 3 dashboards, 46 fine-grained permissions, 235 automated PHP tests across 45 suites. • Operations built in: user-activity reporting, demo-mode AI caps and an instant maintenance switch.

2. Introduction

2.1 Background

Universities depend on an enormous amount of institutional knowledge: course materials, notices, assignment guidelines, attendance records, grading policies and academic regulations. In practice that knowledge is scattered across disconnected platforms. A student may receive a notice on a messaging group, lecture slides through a classroom app, attendance in a spreadsheet they never see, and a transcript only by visiting an academic office in person. The information almost always exists — it is simply unreachable at the moment it is needed.

2.2 Motivation

This project began with a lived problem. A class test had been rescheduled, and the new date was posted in a messaging group the author was no longer part of. The information existed but could not be reached. That is the recurring failure of academic communication: the problem is rarely a lack of information, but its fragmentation across ten disconnected places. UniNexus was built to be the single place institutional knowledge should have lived in all along — and, increasingly, to act on that knowledge rather than merely surface it.

2.3 Objectives

1. Provide students, faculty and administrators with a single unified platform for all academic activity, eliminating the need to switch between disconnected tools.
2. Deliver accurate, hallucination-free AI responses by grounding every answer in verified institutional documents and the user's own materials, with citations and confidence scores.
3. Reduce repetitive faculty and administrative workload by automating quiz and assignment generation, rubric-grading drafts, feedback summarisation, attendance and announcements.
4. Move the assistant from answering to acting, letting students complete real academic tasks in conversation under the same permission checks that govern the UI.
5. Safeguard assessment integrity with layered, privacy-conscious proctoring — on-device camera AI, behaviour logging and risk scoring that produce review signals for humans, never automated verdicts.
6. Enable real-time institutional visibility through centralised analytics, user-activity reporting, AI-usage monitoring, early at-risk detection and governed document management.
7. Build a permission-aware, role-based AI system in which each user accesses only information authorised for their role.
8. Guarantee the system runs anywhere by shipping resilient multi-provider AI with automatic fallback — including a fully offline, keyless provider — and give operators safe public-demo controls.

2.4 Scope

UniNexus covers the complete academic loop for an institution: organising departments, courses, terms and class sections; defining prerequisites and waitlists; enrolling students; publishing materials; setting and grading assignments; running proctored class tests with layered, camera-AI-assisted security; recording attendance; computing grades and transcripts; and surfacing institution-wide analytics alongside operational visitor analytics. A governed document library, together with each user's personal corpus, feeds an AI tutor that answers strictly from authorised sources and can execute a bounded set of academic actions. The platform is architected to scale to thousands of users and to extend to multiple institutions in future work.

3. Problem Analysis

The problems UniNexus addresses were drawn from direct observation of student, faculty and administrative experience. They fall into three groups sharing one root cause: the university has no single, intelligent system of record.

3.1 Student Problems

#	Problem	Impact
1	Fragmented communication across ERP, messaging apps and classroom tools	Constant platform-switching just to stay updated
2	Important messages and files lost in chat history	Resources become unfindable; some are lost entirely
3	No real-time academic visibility (attendance, marks, GPA)	Uncertainty and delayed awareness of standing
4	No centralised way to reach faculty	Questions go unanswered; announcements are missed
5	Own notes and course materials are inert files	Study content cannot be searched, questioned or practised against
6	Manual administrative processes (transcripts, queries)	Time wasted in physical queues for simple information

3.2 Faculty Problems

#	Problem	Impact
1	Manual attendance on paper or spreadsheets	Time-consuming and error-prone for large classes
2	Manual preparation of quizzes, tests and exams	High repetitive workload across multiple sections
3	Manual evaluation, feedback writing and grade calculation	Evenings lost to busywork; slow result preparation
4	No integrity signal on submissions or online tests	Copying and impersonation are invisible unless spotted by chance
5	Struggling students identified too late	Intervention happens after the outcome is already fixed
6	Administrative overload	Less time for actual teaching and mentoring

3.3 Administrative Problems

#	Problem	Impact
1	No real-time visibility into the running semester	Issues surface only after the term ends
2	Difficult cross-department monitoring	Attendance and progress hard to track centrally
3	Ungoverned AI adoption	No control over what the AI may answer from, or what it costs
4	Manual report collection	Data must be gathered from many sources by hand
5	No operational insight into platform usage	Who is visiting, from where and on what devices is unknown

Core problem statement

Students, faculty and administrators all suffer because academic activity is spread across disconnected systems — ERP, messaging groups, classroom apps, spreadsheets, paper and physical offices. This fragmentation produces inefficiency, delay, communication gaps, poor resource management and heavy manual workload. The need is for one intelligent, governed platform.

4. Proposed Solution

UniNexus replaces the fragmented toolset with a single platform offering three role-based experiences — Student, Faculty and Admin — behind one login. The university is organised around a clear academic spine, and an AI layer sits across the whole system, grounded in a governed document library and each user's own materials.

4.1 Academic Structure

The platform is organised top-down. Departments contain courses; courses are offered as class sections each term; students enrol into sections; and everything else — materials, assignments, quizzes, attendance, exams and grades — belongs to the specific section a student is actually in. As a result, each user only ever sees content relevant to their own classes. Registration is two-step (an admin assigns, the student confirms), gated by server-enforced prerequisites, with full sections queuing students on a FIFO waitlist that auto-promotes the head of the queue on any drop. A section's teacher must belong to the same department as the course, so students only ever see and message faculty from their own programme.

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Departments → Courses → Class Sections (per term) → Enrolled Students
 ↓  Materials · Assignments · Quizzes · Attendance · Exams · Grades
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4.2 Three Role-Based Experiences

Role	Purpose	What they primarily do
Student	Learn, ask & act	Ask the streaming AI tutor (grounded in the library and their own notes/materials); let it act — book office hours, generate quizzes and flashcards, plan study; register for sections; submit assignments; take proctored class tests; track attendance, GPA, transcript and concept mastery.
Faculty	Teach & assess	Generate and publish quizzes and assignments; run a shared question bank; take attendance; grade with AI-drafted rubric scores and feedback; review similarity flags and peer ratings; act on at-risk warnings; publish office hours; message their own students; review proctoring dossiers.
Admin	Govern & monitor	Manage users, roles and 46 permissions; departments, terms, courses, sections, prerequisites and waitlists; run the document approval workflow; gate which proctoring layers faculty may use; configure and monitor the AI; view institution-wide analytics and user activity; operate demo caps and the maintenance switch.

4.3 The Pluggable AI Engine

UniNexus's AI runs behind a single provider interface, so the underlying "brain" can be swapped without changing the platform. Four providers are implemented on that contract: **OpenAI** (gpt-4o plus embedding models, native HTTP, no SDK), **OpenRouter** (an OpenAI-wire-compatible chain of free models ending in openrouter/auto), **Jina** (free multilingual embeddings) and a built-in **Mock** provider — deterministic, keyless, always available, and capable of both streaming and tool-shaped requests.

Chat runs through a fallback chain: the admin picks OpenAI or OpenRouter as primary, the other backs it up, and the Mock provider is the terminal, never-fail link. Embeddings resolve separately (with optional dual-embedding so retrieval survives a primary embeddings API dying). Vectors are model-tagged and retrieval filters to the active model, so providers never cross-compare — and switching providers automatically dispatches a re-embedding of the knowledge base. If a live provider is selected but no key is configured, the system falls back automatically rather than failing, so the entire product can be demonstrated with zero external setup or cost.

5. System Architecture

UniNexus is a Laravel + Inertia monolith with a domain-driven service layer over a single MySQL instance. There is no external vector database and no third-party LLM SDK, which keeps the system lean enough to run on modest infrastructure while remaining production-ready. Notably, there is no separate REST API: "API-style" actions are web routes returning Inertia responses, guarded by session auth plus role and permission middleware — a deliberate simplification.

5.1 Layered Overview

Layer	Technology	Responsibility
Presentation	Vue 3 · Inertia 2 · Tailwind · Vite · SSE	Fast, role-differentiated SPA; token-by-token streaming of AI answers and tool events
Application	Role/permission middleware · thin controllers · form requests · policies	Guards every route; validates input; enforces per-record ownership; orchestrates services
Domain (DDD)	User · Academic · Chat · RAG · Search · Notification · Analytics · Community	All business logic, behind interfaces rather than implementations
Infrastructure	AI provider adapters · document storage · text extraction · queued jobs · visit tracking	Swappable adapters; embedding, indexing, screening and evidence pruning run asynchronously
Data	MySQL 8 · embeddings table · audit log · visits log	Structured academic data, document vectors, activity and page-view history in one database

5.2 Technology Choices and Rationale

- **Laravel 11 (PHP 8.2+)** — mature tooling, a queue system for embedding, screening and evidence jobs, and layered authorisation: route middleware → permission-aware form request → per-record policy.
- **Vue 3 + Inertia 2 + Tailwind** — a single-page experience with no separate API client to build or version, with a role-differentiated interface resolved per login.
- **MySQL 8 as both database and vector store** — structured academic data and document embeddings live in the same database; cosine similarity is computed in-application. No separate search engine to deploy, operate or pay for. The vector store is isolated behind a clean swap point for a managed vector DB at scale.
- **Pluggable AI providers** — provider logic lives behind a contract (chat, streaming chat, tool calling, embedding, text extraction), enabling the OpenAI, OpenRouter, Jina and offline Mock providers — and future backends — on the same interface, chained with automatic fallback.
- **Queued jobs** — document processing, personal-corpus sync, submission-similarity screening, email digests and proctoring-evidence pruning run asynchronously, so user-facing requests stay fast.
- **Operational middleware** — a global maintenance-mode handler fronts every request; visit tracking runs after the response is flushed, adding zero latency; and a demo-mode AI limiter meters only the request-consuming AI endpoints.

Key architectural decision

Running vector search natively inside MySQL — instead of adding a dedicated vector database — removes an entire class of infrastructure, cost and operational complexity. Together with the multi-provider fallback chain ending in the keyless Mock provider, this decision is what makes UniNexus both demonstrable on any laptop and deployable on inexpensive hosting.

6. The AI Core

The trustworthiness of UniNexus rests on a simple principle: the AI never answers from memory alone. Every answer is constructed from passages retrieved out of sources the user is authorised to see, and those sources are shown back to them. The pipeline runs entirely in-house.

6.1 The RAG Pipeline

9. **Upload & approve.** An administrator curates the knowledge base; only approved documents may ever be used. Approval, rejection, change requests and comments are all audited.
10. **Chunk.** Each approved document is split into small, overlapping, page-aware passages.
11. **Embed.** Each passage becomes a vector embedding, stored in MySQL alongside its source.
12. **Retrieve.** The user's question is embedded and matched against those vectors by cosine similarity (top-K), scoped to what that user may access.
13. **Ground, stream & cite.** The retrieved passages are supplied to the model as context. The answer streams back token by token with numbered citations, a confidence score and suggested follow-ups; where the sources do not cover something, the AI says so rather than inventing.

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Question → Retrieve authorised passages → Ground → Stream a cited, confidence-scored answer
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6.2 Personal Corpus — "Chat with My Materials"

Retrieval is not limited to the central library. A student's own notes — including handwritten pages transcribed by gpt-4o vision OCR — and the course materials of their enrolled sections are indexed through the same chunk → embed → retrieve → cite pipeline as shadow documents. The retrieval scope for any query is therefore the union of: library documents the user may see, their own notes, and the materials of the sections they are enrolled in or teach. Citations are labelled by origin ("Personal Note", "Course Material"), and a global scope guarantees personal content can never leak into the admin library, approvals, dashboards or analytics.

6.3 Agentic Chat — Tool Calling

The student tutor does more than answer. The provider contract accepts function-calling schemas, and a bounded agent loop (a maximum of three rounds, student chat only) executes requested tools, feeds the results back, and repeats until the model produces a plain answer. Eight tools are exposed: check upcoming deadlines, list my courses, list office-hour slots, book a slot, cancel a booking, generate a practice quiz, generate a flashcard deck, and create a study task.

Safety by construction. Every tool delegates to the real domain service — the same code path the UI uses — so RBAC, section scoping and the atomic office-hours claim bind AI actions exactly as they bind a button click. The tool trail is persisted and rendered step by step in the conversation. A segmented **Agent / Answers-only** switch above the composer puts the student in charge: in Answers-only mode, tool definitions are never attached to the provider call at all — a server-side guarantee, not a UI suggestion.

6.4 Permission-Aware Retrieval

Retrieval respects who may see what. Because the relevance search is filtered by permission and enrolment, two people asking the same question may receive answers built from different sources — a student literally cannot retrieve a passage they are not authorised to see. Any change to the library is reflected immediately: rejecting or editing a document removes its passages from every future answer.

6.5 Why This Beats a Generic Chatbot

Dimension	Generic chatbot	UniNexus
Source of answers	The open internet	Approved institutional documents plus your own notes and course materials
Verifiability	No citations	Citations to document and page, with a confidence score
Permissions	None	Role- and enrolment-aware — users retrieve only what they may see
Governance	None	Documents approved before they ever reach the AI; per-user AI usage monitored and revocable
Action	Text only	Executes real academic tasks through permission-checked tools, with a visible trail
Cost / dependency	Requires a paid API	Falls back across providers to a fully offline, keyless Mock — it can never be down

7. Core Features

7.1 Student Features

- **Streaming AI tutor.** Grounded, cited, confidence-scored answers with selectable modes (academic, exam prep, assignment help, research, career guidance, general) and answer language — drawing on the library, your own notes and your sections' materials.
- **Agentic actions.** Eight in-chat tools with a live activity trail, a hard Agent / Answers-only switch, and an "⚡ Agent" badge on replies where the tutor actually acted.
- **Practice & recall.** Self-serve AI practice quizzes (instant server-graded, missed questions become flashcards), deterministic self-quizzing from the course question bank (no AI required), and flashcards with SM-2 spaced repetition.
- **My Progress & concept mastery.** GPA, attendance, test, assignment and activity trends, plus a per-topic mastery map (blended from class tests, practice accuracy and recall) with one-click "practice this" / "make flashcards" on weak concepts.
- **OCR handwritten notes.** Photograph a page; gpt-4o vision transcribes it into a clean, searchable note that is automatically indexed for the AI tutor.
- **AI Study Planner.** Turns deadlines into a realistic study schedule, saved straight to tasks.
- **Proctored class tests.** Timed, layered-security quizzes — including camera-AI proctoring where enabled — auto-graded the moment they are submitted.
- **Academic tracking.** Live attendance, GPA/CGPA, transcript, degree roadmap, and a unified calendar with .ics export and subscription.
- **Registration.** Confirm assigned sections with prerequisite met/unmet badges and live waitlist positions.
- **Community.** Group study rooms, course discussion feeds, office-hours booking, an opt-in gamified leaderboard, anonymous peer review and anonymous course feedback.

- **Productivity.** ⌘K semantic global search, saved answers, notes, tasks, weekly email digests with deadline nudges, and real-time messaging with faculty.

7.2 Faculty Features

- **AI Teaching Assistant.** Streaming generation of draft quizzes, assignments and rubrics — drafts only, until the teacher publishes.
- **Question bank.** A per-course shared bank of reusable questions: add manually, import from past class tests (duplicates skipped), or spin a selection into a draft test.
- **Layered proctoring.** Pick, per test, from the admin-approved set of 17 security layers — including the on-device camera-AI layers — and review each attempt through a dossier: identity and fingerprint, behaviour timeline, 0–100 risk score with contributing factors, recording playback and a trigger-labelled snapshot photo strip.
- **AI-assisted rubric grading.** "Draft grade with AI" reads the actual submission and pre-fills per-criterion scores with justifications plus a suggested overall grade and feedback — all editable, reviewed and saved by the teacher. Nothing auto-releases.
- **Submission similarity screening.** Submission text (including extracted PDF/DOCX content) is chunked and embedded; high-similarity pairs within an assignment are flagged with matching excerpts and a side-by-side comparison. A review signal, never a verdict.
- **At-risk early warning.** Students flagged live on four signals — attendance, missed deadlines, test average and grade — at High or Watch level, with one-click messaging to intervene.
- **Anonymous course feedback.** Open a mid-semester window per section; results unlock only at three or more responses, with an AI button that summarises themes into Going well / Concerns / Suggestions.
- **Peer review.** An optional per-assignment toggle; average peer ratings surface in the grading workspace.
- **One-click attendance,** section analytics, office-hours slots and discussion moderation in their own sections.

7.3 Administrator Features

- **Document approval & knowledge base.** A review queue where approval automatically extracts, chunks, embeds and indexes a document into the AI's knowledge — and rejection or edit removes it instantly.
- **AI governance.** Provider, models, creativity, passages-per-question, relevance threshold, tone and answer languages are all configurable, with a write-only securely stored API key and a live "Test connection" check; per-user AI usage is monitored and access can be revoked or reinstated.
- **Exam-security gate.** Admins decide, per layer, whether each of the 17 proctoring layers is available to faculty and whether it is on by default — with a built-in "How each layer works" guide that reflects the live configured timings, so documentation can never drift from behaviour.
- **User Activity panel.** Who visited, from where (referrer), on what device, and an IP-derived location — with summary stats, device / top-country / top-page breakdowns, filters and a paginated feed. Logged after the response ships (zero page latency), with configurable retention and a scheduled prune.
- **Demo-mode governance.** In demo mode every account is capped at a fixed number of AI requests across all AI surfaces, so a public demo can't drain the shared provider budget; outside demo mode nothing is metered.
- **Hidden maintenance switch.** A URL-driven operator control (?live=false / ?live=true) puts the whole app behind a maintenance page or flips it back live for everyone; the state persists until toggled, unlock is honoured first so a locked site can always be reopened, and the health check is never gated.
- **Academic governance.** Users, a visual role-and-permission editor with 46 fine-grained permissions and time-limited grants, departments, terms and rollover, courses, sections, prerequisites and waitlists — with a protected admin account that can never be locked out, and a full audit log.

- **Oversight.** Institution-wide analytics and top queries, a discussion moderation queue, announcements, exams and a live system-health monitor for database, storage, jobs and AI status.

7.4 Platform-Wide Capabilities

- **Integrity.** Correct answers never reach the browser during a live attempt; timers are tamper-proof; grading is server-authoritative; every attempt leaves an evidence trail.
- **Layered AI proctoring (17 layers).** Fullscreen, tab-switch detection, clipboard block, one-question-at-a-time, disable-back, question and option randomisation, identity watermark, integrity notice, browser fingerprint, behaviour logging, risk scoring, webcam and screen recording — plus on-device camera AI: a blink-verified **face-liveness gate** (questions stay hidden until a live face is confirmed — a photo can't blink; face loss blurs then locks the paper with escalating warnings and a logged fail-safe bypass), **phone & second-face detection** (a review signal with photo evidence, never an auto-violation), and **snapshot evidence** (photo bursts at flagged moments plus random samples — moment-based evidence instead of hours of video). Detection runs entirely in the browser; no frames are uploaded for it. Recordings and snapshots are stored privately and pruned on a retention schedule.
- **Semantic global search (⌘K).** One search by meaning across documents, notes, materials, courses, assignments, discussions and past AI chats — always scoped to what the user may access.
- **Real-time messaging & presence.** Students and the faculty they share a class with chat live, with presence, typing indicators and unread counts; group study rooms extend the same engine to sections.
- **Notifications & digests.** Grades, materials, assignments, exams and announcements reach exactly the right people; a Monday email digest and daily due-soon nudges keep deadlines visible, with opt-out in settings.

8. Implementation Status & Results

UniNexus is a mature, demonstrable platform. Every core context — User/RBAC, Academic, Chat, RAG, Search, Document, Notification, Analytics and Community — is implemented, wired end-to-end across the three roles, and runnable with no external dependencies.

8.1 By the Numbers

Metric	Value	Meaning
Role dashboards	3	Student, Faculty and Admin — one login, tailored per role
Fine-grained permissions	46	Every route guarded, every action audited; role grants can expire
Agentic chat tools	8	Real academic actions executed in conversation, permission-checked
Chat modes	6	Academic, exam prep, research, assignment help, career guidance, general
Proctoring layers	17	Independent, admin-gated security layers incl. on-device camera AI
At-risk signals	4	Attendance, missed deadlines, test average and grade — live, per section
AI providers	4	OpenAI, OpenRouter, Jina and the keyless Mock — chained with automatic fallback
Automated tests	235	PHP feature tests across 45 suites, plus Vitest JS component tests
External AI dependencies	0	No vector database, no LLM SDK — runs keyless on the Mock provider

8.2 Recently Shipped

The platform has advanced substantially beyond its original MVP. Delivered since the first submission, in successive waves:

- **Copilot depth & connection:** token-by-token streaming, personal-corpus RAG ("chat with my materials"), AI practice quizzes, the at-risk early-warning system, ⌘K semantic global search, group study rooms, office-hours booking, and .ics calendar export and subscribe.
- **Agentic & assessment wave:** agentic tool calling with the Agent / Answers-only switch, AI-assisted rubric grading, submission similarity screening, the concept mastery map with adaptive review, the per-course question bank, email digests and deadline nudges, anonymous peer review and anonymous mid-semester course feedback, course prerequisites and section waitlists.
- **Learning & community wave:** OCR of handwritten notes, the AI Study Planner, flashcards with SM-2 spaced repetition, "My Progress" learning analytics, the opt-in gamified leaderboard, discussion feeds with moderation, and real-time student–faculty messaging.
- **Camera-AI proctoring wave:** the layered exam-security registry grew to 17 layers with webcam/screen recording and three on-device camera-AI layers — the blink-verified face-liveness gate, phone & second-face detection, and snapshot evidence — plus per-attempt risk-scored dossiers with recording playback and snapshot photo strips, and scheduled evidence pruning.
- **Provider-resilience wave:** OpenRouter and Jina joined OpenAI and the Mock provider behind fallback chains for chat and embeddings, with model-tagged vectors, optional dual-embedding and automatic re-embedding on provider switch.
- **Operations & governance wave:** the admin User Activity panel (visitor, referrer, device and IP-derived location reporting with zero request-path latency), the demo-mode AI usage cap across every AI surface, and the hidden URL-driven maintenance switch.

8.3 Expected Outcomes

- A significant reduction in the time students spend searching for academic information — materials, policies, notices and their own notes are all reachable through one cited AI interface and one semantic search.
- Decreased faculty workload on repetitive tasks: generated assessments, AI-drafted rubric grades and feedback, summarised course feedback and one-click attendance return hours to teaching.
- Higher accuracy and trust in academic information, as answers are grounded in approved sources rather than generic internet data and carry citations and confidence.
- Credible remote assessment: evidence-based, human-reviewed proctoring makes online class tests trustworthy without automated punishment.
- Fewer missed deadlines and better study habits, through agentic actions, the study planner, spaced repetition, the mastery map and proactive email nudges.
- Earlier intervention for struggling students, as faculty see live at-risk flags instead of discovering problems at term’s end.
- Improved administrative decision-making through real-time dashboards, analytics, user-activity reporting, AI-usage monitoring and governed document management.

A note on measurement

The outcomes above are structural — each manual workflow is replaced by a single action or an instant, cited answer. Quantified figures (hours saved, adoption) require a departmental pilot, which is the first item on the validation roadmap. No adoption metrics are claimed here that have not been measured.

9. Impact & Feasibility

9.1 Technical Feasibility

UniNexus is production-ready by construction. Its modular, domain-driven architecture supports secure role-based access control, in-database document indexing and maintainable long-term development, verified by an automated suite of 235 PHP feature tests across 45 suites plus JavaScript component tests. The deliberate choice to avoid an external vector database and any third-party LLM SDK lowers both the operational burden and the cost of running the system — and the multi-provider fallback chain ending in the keyless Mock provider means the whole product, including agentic chat, can be demonstrated on any laptop, offline, at zero cost. Because every AI action passes through the same domain services as the UI, adding capability does not widen the security surface, and the operational layer — activity reporting, demo caps and the maintenance switch — makes a public deployment safe to run.

9.2 Societal Impact

In a country like Bangladesh, where universities often struggle with resource fragmentation and inconsistent academic communication, a centralised AI-powered platform can meaningfully raise both the quality and the equity of education. By making institutional knowledge instantly accessible to every student — regardless of the size of their university — and by cutting administrative friction for faculty, UniNexus can improve academic outcomes at institutions of any scale. It also demonstrates how RAG-based and agentic AI can be responsibly deployed in a high-stakes academic environment: every answer cited, every action permission-checked, every AI-drafted grade reviewed by a human before release, and every proctoring signal treated as evidence for human judgement rather than an automated verdict — with the camera AI running on-device so no video frames ever leave the browser for detection.

10. Scalability & Future Work

UniNexus is designed to scale. Its section-based, multi-tenant-ready architecture supports thousands of students, faculty, departments and documents, with an end-of-term rollover that winds classes down, locks in grades and promotes the next semester automatically. Heavy work — embedding, indexing, similarity screening, digests, evidence pruning — is queued rather than inline, and the vector store sits behind a clean swap point so a managed vector database can replace the in-MySQL implementation at scale without touching domain code.

10.1 Roadmap

Stage	Items
Shipped	Streaming, cited RAG tutor · personal-corpus RAG · agentic tool calling (8 tools) · AI quiz generation → layered proctored class tests (17 layers incl. face liveness, phone detection, snapshot evidence) · AI-assisted rubric grading · submission similarity screening · at-risk early warning · concept mastery map · practice quizzes & question bank · flashcards with SM-2 · OCR notes · study planner · ⌘K semantic search · study rooms · office hours · discussions · prerequisites & waitlists · peer review · anonymous course feedback · email digests · real-time messaging · RBAC with audit log · document approval pipeline · AI usage governance · multi-provider fallback (OpenAI, OpenRouter, Jina, Mock) · user-activity reporting · demo-mode AI cap · hidden maintenance switch.
Planned next	Telegram and WhatsApp alerts for assignments, quizzes and announcements · a digital library with an AI assistant answering strictly from its own content.
On hold (designed)	Voice interaction (speech-to-text and text-to-speech) · lecture-audio transcription · completion certificates · expanded real-time presence — intentionally deferred pending additional infrastructure.

Stage	Items
Vision	ML-driven predictive analytics building on the shipped rule-based at-risk signals · additional AI providers (Gemini, self-hosted models) · a managed vector store at scale · document version history and longer conversation memory · multi-institution support.

The long-term vision is to transform universities into AI-driven intelligent ecosystems, where academic knowledge becomes instantly accessible, administrative friction is minimised, and educational productivity is significantly enhanced.

11. Conclusion

UniNexus answers a problem every student and teacher recognises: the information they need exists, but it is scattered and hard to reach. By unifying academic workflows into one role-aware platform, grounding its AI in the institution's own approved documents and the learner's own materials, and letting that AI act — safely, visibly and under the same permissions as a human click — UniNexus delivers answers that are not only fast but verifiable, and help that is not only informative but actionable.

Its citation-backed, permission-aware, governed design — extended now to evidence-based assessment integrity and resilient multi-provider AI — makes it suitable for the high-stakes reality of academic life; its lean, dependency-free architecture makes it genuinely deployable on the infrastructure universities actually have.

One platform. Grounded AI. Every answer, provable.

That is the promise UniNexus is built to keep.